

Study and Comparative Analysis of Public Lithium-Ion Battery Datasets for SOC and SOH Estimation Research

Submitted by:

Shrijan Samrat

Summer Intern, NIT-C

GUIDE: Dr. Nimal Madhu

1. Introduction:

Lithium-ion batteries have become the dominant energy storage technology for electric vehicles, portable electronics, renewable energy storage systems, and smart grid applications due to their high energy density, long cycle life, and high efficiency. The performance and reliability of Battery Management Systems (BMS) strongly depend on accurate estimation of battery parameters such as State of Charge (SOC) and State of Health (SOH).

Modern battery research increasingly relies on publicly available battery datasets for the development, validation, and benchmarking of SOC and SOH estimation algorithms. These datasets provide valuable information such as voltage, current, temperature, capacity degradation, cycle life, and aging behavior under different operating conditions. Researchers use these datasets to develop machine learning models, battery degradation models, Kalman Filter-based estimators, and hybrid intelligent Battery Management Systems.

This report presents a study of publicly available lithium-ion battery datasets commonly used in battery state estimation research. Different datasets are analyzed with respect to battery chemistry, available parameters, aging information, data structure, and suitability for machine learning applications.

2. Dataset Sources:

A variety of publicly available battery datasets have been developed by research institutions, universities, and industrial organizations to support battery modeling, SOC estimation, SOH estimation, degradation analysis, and machine learning research. These datasets differ in battery chemistry, operating conditions, data collection methodology, available parameters, and intended applications.

For this study, ten widely used battery datasets were selected for comparison and analysis. These include datasets developed by NASA, LG Energy Solution, the University of Oxford, CALCE, Stanford University, Panasonic, Toyota, Sandia National Laboratories, Huawei, and Mendeley research repositories. Together,

these datasets provide diverse battery chemistries, aging conditions, and operational characteristics suitable for SOC and SOH estimation research.

Dataset	Organization / Source
NASA Battery Dataset	NASA Ames Research Center
LG Energy Solution Dataset	LG Energy Solution Research
University of Oxford Battery Degradation Dataset	University of Oxford
CALCE Battery Dataset	Center for Advanced Life Cycle Engineering (University of Maryland)
Stanford SEVS Dataset	Stanford University
Panasonic 18650 Dataset	Panasonic Research Dataset
Toyota Battery Dataset	Toyota Research Institute
Sandia Battery Dataset	Sandia National Laboratories
Huawei Battery Aging Dataset	Huawei Research
Mendeley Battery Datasets	Mendeley Data Repository

The datasets listed above represent some of the most widely used public battery datasets for SOC estimation, SOH estimation, battery degradation modeling, and machine learning research. These datasets provide valuable measurements such as voltage, current, temperature, capacity degradation, cycle aging information, and operational profiles under different battery chemistries and testing conditions. They form an important foundation for the development and evaluation of advanced Battery Management Systems and intelligent battery state estimation algorithms.

3. Dataset Comparison Table:

Dataset	Chem.	Volt.	Curr.	Temp.	SOC	SOH	Cycles	Format	App.
NASA Battery Dataset	Li-ion	Yes	Yes	Yes	No	Yes	Yes	MAT	Lab Research
LG Energy Solution Dataset	NMC	Yes	Yes	Yes	Yes	Yes	Yes	CSV	EV Research
University of Oxford Battery Degradation Dataset	NMC	Yes	Yes	Yes	No	Yes	Yes	CSV	Battery Aging
CALCE	Li-ion	Yes	Yes	Yes	No	Yes	Yes	MAT	Battery

Battery Dataset									Aging
Stanford SEVS Dataset	NMC	Yes	Yes	Yes	Yes	Yes	Yes	CSV	Electric Vehicles
Panasonic 18650 Dataset	NCA	Yes	Yes	Yes	No	Yes	Yes	CSV	Cell Characterization
Toyota Battery Dataset	Li-ion	Yes	Yes	Yes	No	Yes	Yes	CSV	EV Research
Sandia Battery Dataset	Li-ion	Yes	Yes	Yes	No	Yes	Yes	CSV	Energy Storage
Huawei Battery Aging Dataset	Li-ion	Yes	Yes	Yes	No	Yes	Yes	CSV	Battery Aging
Mendeleev	Mixed (LFP/N	Yes	Yes	Varies	Varies	Varies	Yes	CSV/MAT	Research

Battery Datasets	MC/LC O)								Reposito ry
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Table 1: Comparison of Publicly Available Battery Datasets for SOC and SOH Estimation Research

4. Chemistry-wise Dataset Analysis 📱⚡

Lithium-ion battery datasets used for SOC and SOH estimation research contain different battery chemistries, each exhibiting unique electrochemical, thermal, and voltage characteristics. These chemistry-dependent behaviors significantly influence battery modeling, state estimation accuracy, degradation analysis, and machine learning performance.

The most common battery chemistries represented in public battery datasets include Lithium Iron Phosphate (LFP), Nickel Manganese Cobalt (NMC), Lithium Cobalt Oxide (LCO), Nickel Cobalt Aluminum Oxide (NCA), and Nickel Metal Hydride (NiMH). Understanding these chemistries is essential because SOC estimation methods often depend on chemistry-specific voltage behavior and degradation patterns.

4.1 LFP (Lithium Iron Phosphate)

LFP batteries are widely used in electric vehicles and energy storage systems because of their excellent thermal stability, long cycle life, and improved safety characteristics. One important characteristic of LFP

batteries is the presence of a very flat voltage plateau over a large SOC range. While this improves operational stability, it makes voltage-based SOC estimation more challenging because large SOC variations correspond to relatively small voltage changes.

4.2 NMC (Nickel Manganese Cobalt)

NMC batteries are among the most commonly used lithium-ion chemistries in electric vehicles due to their high energy density and balanced performance. Unlike LFP batteries, NMC batteries exhibit a more gradual voltage variation with SOC, making voltage-based SOC estimation easier. However, NMC batteries generally have lower thermal stability compared to LFP batteries and require careful Battery Management System monitoring.

4.3 LCO (Lithium Cobalt Oxide)

Lithium Cobalt Oxide (LCO) batteries are widely used in portable electronic devices such as smartphones, laptops, and tablets. LCO batteries offer high energy density but relatively lower cycle life and thermal stability compared to other lithium-ion chemistries. Their voltage-SOC characteristics are moderately nonlinear and are commonly used in battery estimation research.

4.4 NCA (Nickel Cobalt Aluminum Oxide)

NCA batteries provide very high energy density and are commonly used in high-performance electric vehicles. They offer excellent energy storage capability but require advanced thermal management and monitoring due to safety considerations. NCA datasets are particularly useful for studying advanced battery degradation and SOC estimation techniques.

4.5 NiMH (Nickel Metal Hydride)

NiMH batteries were widely used before the adoption of modern lithium-ion batteries and are still found in certain hybrid vehicle applications. Although they have lower energy density compared to lithium-ion batteries, they offer good safety and reliability characteristics. NiMH datasets provide useful benchmark information for comparative battery research.

Chemistry	OCV Behavior	Voltage Plateau	Thermal Stability	SOC Estimation Difficulty
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LFP	Flat	Strong	Excellent	High
NMC	Nonlinear	Moderate	Good	Moderate
LCO	Nonlinear	Moderate	Moderate	Moderate
NCA	Nonlinear	Weak	Moderate	Moderate
NiMH	Gradual	Weak	Good	Low

Table 2: Comparison of Battery Chemistry Characteristics Relevant to SOC Estimation

The diversity of battery chemistries available in public datasets is important for developing robust SOC and SOH estimation algorithms. Since voltage behavior, thermal characteristics, and degradation mechanisms vary significantly among chemistries, machine learning models trained on one chemistry may not generalize effectively to another. This remains an active research challenge in multi-chemistry battery intelligence systems.

5. SOC-Relevant Parameters

State of Charge (SOC) estimation relies on several measurable battery parameters that provide information about the available charge remaining inside a battery. Public battery datasets typically contain electrical and

thermal measurements that are used as inputs for classical estimation methods, Kalman Filter algorithms, and machine learning models.

Among all available battery measurements, voltage, current, temperature, internal resistance, and Open Circuit Voltage (OCV) are the most important parameters used for SOC estimation. The quality, sampling frequency, and availability of these parameters strongly influence the accuracy of battery state estimation algorithms.

5.1 Voltage

Voltage is one of the most important parameters used for battery SOC estimation. Battery voltage varies with SOC and battery chemistry, making it a primary indicator of battery charge state. Voltage measurements are extensively used in OCV-based methods, Kalman Filters, and machine learning models.

5.2 Current

Current measurements are essential for Coulomb Counting and dynamic battery state estimation methods. Battery charging and discharging currents directly affect SOC variation and provide important information regarding battery operating conditions.

5.3 Temperature

Temperature significantly influences battery electrochemical behavior, internal resistance, charging efficiency, and voltage response. Including temperature information improves SOC estimation accuracy, particularly under varying environmental conditions.

5.4 Internal Resistance

Internal resistance changes with SOC, temperature, battery aging, and operating conditions. Resistance measurements provide valuable information about battery condition and can improve the performance of advanced estimation algorithms.

5.5 Open Circuit Voltage (OCV)

Open Circuit Voltage represents the stabilized battery voltage measured under no-load conditions. OCV has a strong relationship with SOC and is widely used in battery calibration, lookup-table methods, and model-based estimation techniques.

Parameter	Importance for SOC Estimation
Voltage	Indicates battery charge state

Current	Tracks charge and discharge behavior
Temperature	Captures thermal effects
Internal Resistance	Reflects battery condition
OCV	Provides SOC reference under resting conditions

Table 3: Important Parameters Used for SOC Estimation

The availability of voltage, current, temperature, resistance, and OCV measurements makes public battery datasets valuable resources for SOC estimation research. These parameters serve as primary inputs for both traditional estimation techniques and modern machine learning models.

6. SOH-Relevant Parameters

State of Health (SOH) represents the overall health condition and degradation status of a battery relative to its original condition. Unlike SOC estimation, which focuses on the remaining charge available at a given moment, SOH estimation evaluates long-term battery degradation, performance loss, and remaining service capability.

Public battery datasets often include aging-related measurements that allow researchers to study degradation mechanisms and develop predictive SOH estimation models. Parameters such as capacity fade, internal resistance growth, cycle count, and temperature variation are commonly used for battery health assessment.

6.1 Capacity Fade

Capacity fade refers to the gradual reduction in the maximum charge that a battery can store as it ages. Repeated charging and discharging cycles cause electrochemical degradation, reducing the available battery capacity over time. Capacity fade is one of the most widely used indicators of battery health.

6.2 Internal Resistance Growth

As batteries age, their internal resistance gradually increases due to electrochemical and structural changes within the cell. Increased resistance leads to higher energy losses, reduced efficiency, and degraded battery performance. Internal resistance growth is therefore an important SOH indicator.

6.3 Cycle Count

Cycle count represents the number of complete charging and discharging cycles experienced by a battery. Battery degradation generally increases with cycle count, making this parameter useful for studying aging behavior and predicting remaining battery life.

6.4 Temperature Variation

Temperature significantly affects battery aging and degradation. High operating temperatures accelerate capacity loss and internal resistance growth, while extremely low temperatures can reduce battery efficiency and performance. Temperature data is therefore essential for accurate SOH estimation.

Parameter	Importance for SOH Estimation
Capacity Fade	Direct indicator of battery degradation
Internal Resistance Growth	Indicates aging and efficiency loss
Cycle Count	Tracks battery usage history
Temperature Variation	Influences degradation rate

Table 4: Important Parameters Used for SOH Estimation

SOH estimation depends on long-term battery degradation indicators rather than short-term operational measurements. Parameters such as capacity fade, resistance growth, cycle count, and temperature variation provide valuable information for predicting battery aging, maintenance requirements, and remaining useful life.

7. ML Suitability Analysis

The increasing use of machine learning and deep learning techniques for battery SOC and SOH estimation has made dataset quality a critical factor in battery research. Public battery datasets serve as the foundation for training, validating, and benchmarking machine learning models. The suitability of a dataset for machine learning applications depends on factors such as data volume, measurement quality, label availability, operating diversity, and real-world relevance.

Datasets containing voltage, current, temperature, aging information, SOC labels, and SOH indicators are particularly valuable because they enable supervised learning approaches and advanced battery intelligence applications. Therefore, evaluating dataset suitability is essential before developing machine learning-based estimation models.

7.1 Data Size

Large datasets generally improve machine learning performance because they expose models to a wider range of operating conditions, battery behaviors, and degradation patterns. Datasets containing extensive charging and discharging cycles are especially useful for training robust SOC and SOH estimation models.

7.2 Noise and Measurement Quality

Battery measurements collected from sensors often contain noise, fluctuations, calibration errors, and environmental disturbances. Excessive noise can reduce machine learning accuracy and increase model uncertainty. High-quality datasets with reliable sensor measurements improve model training and prediction performance.

7.3 Missing Values

Missing values are common in real-world battery datasets due to communication failures, sensor malfunctions, and incomplete data collection processes. Datasets with fewer missing values are generally preferred because they require less preprocessing and provide more reliable training data.

7.4 Label Availability and Quality

Machine learning models require accurate target labels for supervised learning. SOC labels, SOH labels, capacity measurements, and degradation indicators are essential for training estimation models. Datasets with well-defined and accurately measured labels are more suitable for machine learning applications.

7.5 Real-Time Applicability

Datasets collected under realistic operating conditions are highly valuable for practical Battery Management System development. Real-world datasets containing dynamic load variations, temperature fluctuations, and aging behavior help machine learning models generalize better to practical deployment scenarios.

Evaluation Criterion	Importance for ML Models
Data Size	Improves model learning and generalization
Noise Level	Affects prediction accuracy
Missing Values	Influences data reliability
Label Quality	Essential for supervised learning
Real-Time Applicability	Improves practical deployment performance

Table 5: Factors Affecting Dataset Suitability for Machine Learning Applications

Dataset	ML Suitability	Reason
NASA	High	Rich aging and degradation data
LG Energy Solution	High	SOC/SOH relevant measurements
Oxford	High	Extensive degradation cycles
CALCE	High	Battery aging and health studies
Stanford SEVS	Very High	EV-focused operational data
Panasonic 18650	Moderate	Cell-level characterization
Toyota	High	EV research relevance
Sandia	Moderate	Energy storage focus
Huawei	High	Aging analysis capability
Mendeley	Variable	Depends on dataset source

Table 6: Machine Learning Suitability of Public Battery Datasets

Among the datasets surveyed, NASA, CALCE, Oxford, Stanford SEVS, and LG Energy Solution datasets appear particularly suitable for machine learning-based SOC and SOH estimation research due to their rich measurement data, aging information, and relevance to real-world battery applications. These datasets provide valuable resources for developing advanced machine learning, deep learning, and TinyML-based battery intelligence systems.

8. Challenges in Dataset Usage ⚠️

Although publicly available battery datasets have significantly accelerated SOC and SOH estimation research, several challenges continue to limit their effectiveness for developing robust and universally applicable battery intelligence systems. Differences in battery chemistry, operating conditions, data collection methodologies, labeling strategies, and degradation patterns often create difficulties when comparing datasets or training generalized machine learning models.

Understanding these limitations is important because dataset quality directly influences estimation accuracy, model reliability, and real-world deployment capability.

8.1 Battery Chemistry Diversity

Different battery datasets contain different chemistries such as LFP, NMC, LCO, and NCA. Since each chemistry exhibits unique voltage-SOC characteristics, degradation behavior, and thermal responses, machine learning models trained on one chemistry may perform poorly on another.

8.2 Inconsistent Data Formats

Public battery datasets are available in multiple formats including CSV, MAT, Excel, and proprietary structures. This lack of standardization increases preprocessing effort and complicates cross-dataset analysis.

8.3 Missing and Noisy Data

Many battery datasets contain missing values, sensor noise, measurement errors, and incomplete experimental records. Such issues can negatively affect machine learning training and reduce estimation accuracy.

8.4 Limited SOC and SOH Labels

Several public datasets provide raw measurements such as voltage and current but do not directly include SOC or SOH labels. Researchers must often compute or estimate these labels before model development.

8.5 Limited Real-World Operating Conditions

Many datasets are collected under controlled laboratory conditions. As a result, machine learning models trained on such data may struggle when deployed in real-world environments with varying temperatures, dynamic loads, and unpredictable usage patterns.

Challenge	Impact on Research
Chemistry Diversity	Reduces model generalization
Data Format Differences	Increases preprocessing effort
Missing Values	Reduces data quality
Limited Labels	Restricts supervised learning
Laboratory Conditions	Limits real-world applicability

Table 7: Major Challenges Associated with Public Battery Datasets

9. Research Gaps

Despite the availability of numerous battery datasets and estimation techniques, several important research gaps remain in battery intelligence research. Addressing these gaps is essential for developing accurate,

scalable, and deployable SOC and SOH estimation systems suitable for future electric vehicles and energy storage applications.

9.1 Multi-Chemistry Estimation

Most existing estimation models are developed and validated using a single battery chemistry. Research on generalized multi-chemistry SOC and SOH estimation remains limited and represents an important future research direction.

9.2 Physics-Informed Machine Learning

Many machine learning models operate as black-box systems without incorporating battery physics. Combining electrochemical knowledge with machine learning remains an active area of research for improving interpretability and reliability.

9.3 Lightweight TinyML Models

Although deep learning models provide excellent estimation accuracy, their computational requirements often exceed the capabilities of embedded Battery Management Systems. Developing lightweight TinyML models remains a significant research challenge.

9.4 Cross-Dataset Generalization

Models trained on one dataset frequently experience performance degradation when evaluated on different datasets. Improving cross-dataset generalization remains an important requirement for practical deployment.

9.5 Real-World Battery Data

There is a growing need for publicly available datasets collected under realistic operating conditions, including electric vehicle driving cycles, environmental variations, and long-term field operation.

Research Gap	Future Importance
Multi-Chemistry Estimation	High
Physics-Informed ML	High
TinyML Deployment	Very High
Cross-Dataset Generalization	High
Real-World Data Collection	Very High

Table 8: Major Research Gaps in Battery State Estimation Research

10. Conclusion

This report presented a comparative study of publicly available battery datasets commonly used for SOC and SOH estimation research. Ten widely used datasets were examined with respect to battery chemistry, available measurements, aging information, data formats, and suitability for machine learning applications.

The analysis showed that battery datasets differ significantly in chemistry, operational conditions, data quality, and available labels. Parameters such as voltage, current, temperature, capacity fade, and internal resistance play critical roles in battery state estimation and machine learning model development.

Among the surveyed datasets, NASA, CALCE, Oxford, Stanford SEVS, and LG Energy Solution datasets were found to be particularly valuable for battery intelligence research due to their rich operational and degradation information. However, challenges such as chemistry diversity, missing data, limited labels, and lack of standardized formats continue to affect battery estimation research.

Future work is expected to focus on multi-chemistry estimation, physics-informed machine learning, TinyML deployment, and the development of robust battery intelligence systems capable of operating across diverse real-world conditions. These advances will contribute significantly to the next generation of Battery Management Systems, electric vehicles, and intelligent energy storage technologies.

11. References

- [1] NASA Ames Research Center, “NASA Battery Dataset,” Prognostics Center of Excellence (PCoE), NASA.
- [2] CALCE Battery Research Group, “Battery Data Sets,” Center for Advanced Life Cycle Engineering (CALCE), University of Maryland.
- [3] University of Oxford, “Battery Degradation Dataset for Lithium-Ion Batteries,” Oxford Battery Intelligence Laboratory.
- [4] LG Energy Solution Research, “Lithium-Ion Battery Data and Performance Studies.”
- [5] Stanford University, “SEVS Battery Dataset for Electric Vehicle Applications.”
- [6] G. L. Plett, Battery Management Systems, Volume I: Battery Modeling, Artech House, 2015.
- [7] M. A. Hannan, M. M. Hoque, A. Mohamed, and A. Ayob, “Review of State of Charge Estimation Methods for Lithium-Ion Batteries,” IEEE Access, vol. 6, pp. 18356–18375, 2018.
- [8] H. He, R. Xiong, and J. Fan, “Evaluation of Lithium-Ion Battery Equivalent Circuit Models for State of Charge Estimation,” Applied Energy, vol. 88, no. 11, pp. 4138–4146, 2011.

[9] D. Pietro and A. Aniballi, “TinyML-Based Battery State of Charge Estimation on Edge Devices,” IEEE Embedded Systems Letters, 2023.